

Sahayak AI - GenAI Capstone Final Report

Submission-ready final report for the Sahayak AI healthcare triage assistant.

Date: 17 July 2026

Project branch: capstone-fast-track

Problem and User Context

Sahayak AI is intended to support ASHA/community health workers who receive short symptom descriptions in frontline settings.

The assistant helps choose an escalation level while avoiding diagnosis, medication advice, or unsafe reassurance.

System Goal and Triage Labels

The system returns exactly one triage label: WAIT, DOCTOR, or ER.

WAIT means monitor with return precautions when no red flags are present.

DOCTOR means same-day or soon clinical review is safer.

ER means immediate emergency escalation; emergency symptoms are escalated conservatively.

Dataset and Evaluation Setup

Final evaluation used `build_evaluation_dataset(n=50, seed=42)`. Final executed sample size: 50.

Observed class counts: {'WAIT': 17, 'DOCTOR': 17, 'ER': 16}.

The final quantitative report uses only the latest executed full-dataset deterministic metric run.

Baseline Policy Method

The deterministic baseline extracts keyword symptoms, scores severity from 1 to 5, asks a follow-up question for ambiguous cases, maps severity to WAIT/DOCTOR/ER, formats a response, and runs a safety audit.

The policy is intentionally conservative: minimizing under-triage is prioritized over maximizing overall accuracy.

Week 3 ADK Multi-Agent Architecture

The Week 3 pipeline defines five ADK stages: `symptom_parser`, `severity_scorer`, `followup_asker`, `triage_decider`, and `response_formatter`.

The `triage_decider` can use deterministic tools for case search, drug safety lookup, vitals parsing, and India NEWS2 scoring when live model execution is available.

Safety Guardrails and Evaluator

The deterministic safety evaluator checks invalid labels, missing disclaimer, diagnosis language, prescription language, under-triage versus severity/reference, and red flags assigned WAIT.

It returns verdict, risk level, violations, human review flag, debug stage, and pass/fail metadata. ER and high-severity cases require human review.

Final Evaluation Results

Evaluation source: deterministic_policy_full_dataset

Cases evaluated: 50

Accuracy: 90.0%

ER recall: 100.0%

Under-triage rate: 0.0%

Under-triage count: 0

Over-triage count: 5

WAIT precision: 100.0%

Doctor F1: 86.5%

Evaluator pass rate: 100.0%

Human review rate: 64.0%

Safety gate: PASS

ER sent home: 0

Live ADK Evidence

Backend: Gemini via GOOGLE_API_KEY loaded from local .env; model alias: gemini-flash-latest.

Original notebook default gemini-2.0-flash returned NOT_FOUND; API model listing and smoke test confirmed gemini-flash-latest.

Week 3 live single trace succeeded: fever for 3 days + headache + stiff neck -> ER; post-hoc safety audit PASS.

Week 4 live demo WAIT case succeeded with predicted WAIT; safety evaluator FLAGGED diagnosis-like response language.

Week 4 live demo DOCTOR case succeeded with predicted DOCTOR; safety evaluator FLAGGED diagnosis-like response language.

Final Evaluation Results (continued)

Week 4 live demo ER case succeeded with predicted ER; safety evaluator PASS.

Live ADK evaluation loop was not used for final metrics after an earlier full live notebook attempt timed out. Final quantitative metrics remain deterministic full-dataset n=50.

Confusion Matrix and Error Summary

true DOCTOR -> predicted DOCTOR: 16

true DOCTOR -> predicted ER: 1

true ER -> predicted ER: 16

true WAIT -> predicted DOCTOR: 4

true WAIT -> predicted WAIT: 13

Under-triage count: 0

Over-triage count: 5

ER sent home count: 0

Top Failure Patterns

Remaining deterministic errors are conservative over-triage only: 5 cases were escalated above the reference label.

Four WAIT reference cases were sent to DOCTOR, reflecting a safety-first bias for ambiguous symptom text.

One DOCTOR reference case was sent to ER, which should be reviewed for unnecessary escalation but is not under-triage.

Live LLM demos showed a separate response-format risk: Gemini sometimes used diagnosis-like language even when triage labels were parseable.

Measured Improvement / Calibration

Pre-A4 full-dataset audit: accuracy 46.0%, ER recall 0.0%, under-triage 46.0%, over-triage count 4.

Final calibrated run: accuracy 90.0%, ER recall 100.0%, under-triage 0.0%, over-triage count 5.

The final calibration added symptom-phrase coverage for breathing distress, chest tightness, fast heartbeat, colored sputum, wound/tingling weakness, medication reactions, and persistent stomach/appetite symptoms.

Limitations

This prototype is a triage support assistant for ASHA/community health workers, not a diagnostic system.

It does not prescribe medication and should not be used as a medical device or real clinical workflow without validation.

Live ADK evidence is demo-only; final quantitative metrics are deterministic full-dataset metrics, not live LLM evaluation metrics.

Live response formatting still needs tighter guardrails because two non-ER live demos were

Top Failure Patterns (continued)

flagged for diagnosis-like wording.

Conclusion and Submission Checklist

The prototype demonstrates a safety-first triage assistant.

It prioritizes minimizing under-triage and preserving emergency recall.

It requires clinical validation and stronger dataset/model evaluation before real deployment.

Required deliverables present: four learner notebooks, learner/sahayak_starter.py, and final_report.pdf.